

Espaços de Probabilidade

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Resumo

Introduzimos o conceito de espaços de probabilidade e apresentamos as provas das propriedades elementares de uma medida de probabilidade.

Events and probabilities

Summary. The very basic principles and tools of probability theory are set out. An event involving randomness may be described in mathematical terms as a probability space. Following an account of the properties of probability spaces, the concept of conditional probability is explained, and also that of the independence of events. There are many worked examples of calculations of probabilities.

1. Experiments with chance

Many actions have outcomes which are largely unpredictable in advance—tossing a coin and throwing a dart are simple examples. Probability theory is about such actions and their consequences. The mathematical theory starts with the idea of an *experiment* (or *trial*), being a course of action whose consequence is not predetermined. This experiment is reformulated as a mathematical object called a *probability space*. In broad terms, the probability space corresponding to a given experiment comprises three items:

- (i) the set of all possible outcomes of the experiment,
- (ii) a list of all the events which may possibly occur as consequences of the experiment,
- (iii) an assessment of the likelihoods of these events.

For example, if the experiment is the throwing of a fair six-sided die, then the probability space amounts to the following:

- (i) the set $\{1, 2, 3, 4, 5, 6\}$ of possible outcomes,
- (ii) a list of events such as

- the result is 3,
- the result is at least 4,
- the result is a prime number,

(iii) each number 1, 2, 3, 4, 5, 6 is equally likely to be the result of the throw.

Given any experiment involving chance, there is a corresponding probability space, and the study of such spaces is called *probability theory*. Next, we shall see how to construct such spaces more explicitly.

2. Outcomes and events

We use the letter $\mathcal{E}$ to denote a particular experiment whose outcome is not completely predetermined. The first thing which we do is to make a list of all the possible outcomes of $\mathcal{E}$.

The set of all such possible outcomes is called the *sample space* of $\mathcal{E}$ and we usually denote it by Ω . The Greek letter ω denotes a typical member of Ω , and we call each member ω an *elementary event*.

If, for example, $\mathcal{E}$ is the experiment of throwing a fair die once, then

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

There are many questions which we may wish to ask about the actual outcome of this experiment (questions such as “is the outcome a prime number?”), and all such questions may be rewritten in terms of subsets of Ω (the previous question becomes “does the outcome lie in the subset $\{2, 3, 5\}$ of Ω ?”). The second thing which we do is to make a list of all the events which are interesting to us. This list takes the form of a collection of subsets of Ω , each such subset A representing the event “the outcome of $\mathcal{E}$ lies in A ”. Thus we ask “which possible events are interesting to us”, and then we make a list of the corresponding subsets of Ω . This relationship between *events* and *subsets* is very natural, especially because two or more events combine with each other in just the same way as the corresponding subsets combine. For example, if A and B are subsets of Ω , then

- the set $A \cup B$ corresponds to the event “either A or B occurs”,
- the set $A \cap B$ corresponds to the event “both A and B occur”,
- the complement $A^c := \Omega \setminus A$ corresponds to the event “ A does not occur”,¹

where we say that a subset C of Ω “occurs” whenever the outcome of $\mathcal{E}$ lies in C . Thus all set-theoretic statements and combinations may be interpreted in terms of events. For example, the formula

$$\Omega \setminus (A \cap B) = (\Omega \setminus A) \cup (\Omega \setminus B)$$

may be read as “if A and B do not both occur, then either A does not occur or B does not occur”. In a similar way, if $A_1, A_2, \dots$ are events, then the sets $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ represent the events “ A_i occurs, for some i ” and “ A_i occurs, for every i ”, respectively.

¹For any subset A of Ω , the *complement* of A is the set of all members of Ω which are not members of A . We denote the complement of A by either $\Omega \setminus A$ or A^c , depending on the context.

Thus we write down a collection $\mathcal{F} = \{A_i : i \in I\}$ of subsets of Ω which are interesting to us; each $A \in \mathcal{F}$ is called an *event*. In simple cases, such as the die-throwing example above, we usually take $\mathcal{F}$ to be the set of *all* subsets of Ω (called the *power set* of Ω), but for reasons which may be appreciated later there are many circumstances in which we take $\mathcal{F}$ to be a very much smaller collection than the entire power set.² In all cases we demand a certain consistency of $\mathcal{F}$, in the following sense. If $A, B, C, \dots \in \mathcal{F}$, we may reasonably be interested also in the events “ A does *not* occur” and “at least one of $A, B, C, \dots$ occurs”. With this in mind, we require that $\mathcal{F}$ satisfy the following definition.

Definição 1 (1.1). *The collection $\mathcal{F}$ of subsets of the sample space Ω is called an **event space** if*

$$\mathcal{F} \text{ is non-empty,} \tag{1.2}$$

$$\text{if } A \in \mathcal{F} \text{ then } \Omega \setminus A \in \mathcal{F}, \tag{1.3}$$

$$\text{if } A_1, A_2, \dots \in \mathcal{F} \text{ then } \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}. \tag{1.4}$$

We speak of an event space $\mathcal{F}$ as being “closed under the operations of taking complements and countable unions”. Here are some elementary consequences of axioms (1.2)–(1.4).

- (a) An event space $\mathcal{F}$ must contain the empty set $\emptyset$ and the whole set Ω . This holds as follows. By (1.2), there exists some $A \in \mathcal{F}$. By (1.3), $A^c \in \mathcal{F}$. We set $A_1 = A$, $A_i = A^c$ for $i \geq 2$ in (1.4), and deduce that $\mathcal{F}$ contains the union $\Omega = A \cup A^c$. By (1.3) again, the complement $\Omega \setminus \Omega = \emptyset$ lies in $\mathcal{F}$ also.
- (b) An event space is closed under the operation of *finite* unions, as follows. Let $A_1, A_2, \dots, A_m \in \mathcal{F}$, and set $A_i = \emptyset$ for $i > m$. Then $A := \bigcup_{i=1}^m A_i$ satisfies $A = \bigcup_{i=1}^{\infty} A_i$, so that $A \in \mathcal{F}$ by (1.4).
- (c) The third condition (1.4) is written in terms of *unions*. An event space is also closed under the operations of taking finite or countable *intersections*. This follows by the elementary formula $(A \cap B)^c = A^c \cup B^c$, and its extension to finite and countable families.

Here are some examples of pairs $(\Omega, \mathcal{F})$ of sample spaces and event spaces.

Exemplo 2 (1.5). Ω is any non-empty set and $\mathcal{F}$ is the power set of Ω .

Exemplo 3 (1.6). Ω is any non-empty set and $\mathcal{F} = \{\emptyset, A, \Omega \setminus A, \Omega\}$, where A is a given non-trivial subset of Ω .

Exemplo 4 (1.7). $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $\mathcal{F}$ is the collection

$$\emptyset, \{1, 2\}, \{3, 4\}, \{5, 6\}, \{1, 2, 3, 4\}, \{3, 4, 5, 6\}, \{1, 2, 5, 6\}, \Omega$$

of subsets of Ω . This event space is unlikely to arise naturally in practice.

In the following exercises, Ω is a set and $\mathcal{F}$ is an event space of subsets of Ω .

Exercício 1 (1.8). If $A, B \in \mathcal{F}$, show that $A \cap B \in \mathcal{F}$.

²This is explained in Footnote 3 on p. 6.

Exercício 2 (1.9). The *difference* $A \setminus B$ of two subsets A and B of Ω is the set $A \cap (\Omega \setminus B)$ of all points of Ω which are in A but not in B . Show that if $A, B \in \mathcal{F}$, then $A \setminus B \in \mathcal{F}$.

Exercício 3 (1.10). The *symmetric difference* $A \Delta B$ of two subsets A and B of Ω is defined to be the set of points of Ω which are in either A or B but not both. If $A, B \in \mathcal{F}$, show that $A \Delta B \in \mathcal{F}$.

Exercício 4 (1.11). If $A_1, A_2, \dots, A_m \in \mathcal{F}$ and k is a positive integer, show that the set of points in Ω which belong to exactly k of the A_i belongs to $\mathcal{F}$ (the previous exercise is the case when $m = 2$ and $k = 1$).

Exercício 5 (1.12). Show that, if Ω is a finite set, then $\mathcal{F}$ contains an even number of subsets of Ω .

3. Probabilities

From our experiment $\mathcal{E}$, we have so far constructed a sample space Ω and an event space $\mathcal{F}$ associated with $\mathcal{E}$, but there has been no mention yet of probabilities. The third thing which we do is to allocate probabilities to each event in $\mathcal{F}$, writing $\mathbb{P}(A)$ for the probability of the event A . We shall assume that this can be done in such a way that the probability function $\mathbb{P}$ satisfies certain intuitively attractive conditions:

- (a) each event A in the event space has a probability $\mathbb{P}(A)$ satisfying $0 \leq \mathbb{P}(A) \leq 1$,
- (b) the event Ω , that “something happens”, has probability 1, and the event $\emptyset$, that “nothing happens”, has probability 0,
- (c) if A and B are disjoint events (in that $A \cap B = \emptyset$), then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.

We collect these conditions into a formal definition as follows.³

Definição 5 (1.13). A mapping $\mathbb{P} : \mathcal{F} \rightarrow \mathbb{R}$ is called a *probability measure* on $(\Omega, \mathcal{F})$ if

- (a) $\mathbb{P}(A) \geq 0$ for $A \in \mathcal{F}$,
- (b) $\mathbb{P}(\Omega) = 1$ and $\mathbb{P}(\emptyset) = 0$,
- (c) if $A_1, A_2, \dots$ are disjoint events in $\mathcal{F}$ (in that $A_i \cap A_j = \emptyset$ whenever $i \neq j$) then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i). \quad (1.14)$$

We emphasize that a probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ is defined only on those subsets of Ω which lie in $\mathcal{F}$. Here are two notes about probability measures.

- (i) The second part of condition (b) is superfluous in the above definition. To see this, define the disjoint events $A_1 = \Omega$, $A_i = \emptyset$ for $i \geq 2$. By condition (c),

$$\mathbb{P}(\Omega) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \mathbb{P}(\Omega) + \sum_{i=2}^{\infty} \mathbb{P}(\emptyset).$$

³This is where the assumptions of an event space come to the fore. Banach and Kuratowski proved in 1929 that there exists no probability measure $\mathbb{P}$ defined on all subsets of the interval $[0, 1]$ satisfying $\mathbb{P}(\{x\}) = 0$ for every singleton $x \in [0, 1]$.

- (ii) Condition (c) above is expressed as saying that $\mathbb{P}$ is *countably additive*. The probability measure $\mathbb{P}$ is also *finitely additive* in that

$$\mathbb{P}\left(\bigcup_{i=1}^m A_i\right) = \sum_{i=1}^m \mathbb{P}(A_i)$$

for disjoint events A_i . This is deduced from condition (c) by setting $A_i = \emptyset$ for $i > m$.

Condition (1.14) requires that the probability of the union of a countable collection of disjoint sets is the sum of the individual probabilities.⁴

Exemplo 6 (1.15). Let Ω be a non-empty set and let A be a proper, non-empty subset of Ω (so that $A \neq \emptyset, \Omega$). If $\mathcal{F}$ is the event space $\{\emptyset, A, \Omega \setminus A, \Omega\}$, then all probability measures on $(\Omega, \mathcal{F})$ have the form

$$\begin{aligned}\mathbb{P}(\emptyset) &= 0, & \mathbb{P}(A) &= p, \\ \mathbb{P}(\Omega \setminus A) &= 1 - p, & \mathbb{P}(\Omega) &= 1,\end{aligned}$$

for some p satisfying $0 \leq p \leq 1$. $\triangle$

Exemplo 7 (1.16). Let $\Omega = \{\omega_1, \omega_2, \dots, \omega_N\}$ be a finite set of exactly N points, and let $\mathcal{F}$ be the power set of Ω . It is easy to check that the function $\mathbb{P}$ defined by

$$\mathbb{P}(A) = \frac{1}{N}|A| \quad \text{for } A \in \mathcal{F}$$

is a probability measure on $(\Omega, \mathcal{F})$.⁵ $\triangle$

Exercício 6 (1.17). Let $p_1, p_2, \dots, p_N$ be non-negative numbers such that $p_1 + p_2 + \dots + p_N = 1$, and let $\Omega = \{\omega_1, \omega_2, \dots, \omega_N\}$, with $\mathcal{F}$ the power set of Ω , as in Example 1.16. Show that the function $\mathbb{Q}$ given by

$$\mathbb{Q}(A) = \sum_{i: \omega_i \in A} p_i \quad \text{for } A \in \mathcal{F}$$

is a probability measure on $(\Omega, \mathcal{F})$. Is $\mathbb{Q}$ a probability measure on $(\Omega, \mathcal{F})$ if $\mathcal{F}$ is not the power set of Ω but merely some event space of subsets of Ω ?

4. Probability spaces

We combine the previous ideas in a formal definition.

Definição 8 (1.18). A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ of objects such that

- (a) Ω is a non-empty set,
- (b) $\mathcal{F}$ is an event space of subsets of Ω ,
- (c) $\mathbb{P}$ is a probability measure on $(\Omega, \mathcal{F})$.

⁴A set S is called *countable* if it may be put in one-one correspondence with a subset of the natural numbers $\{1, 2, 3, \dots\}$.

⁵The cardinality $|A|$ of a set A is the number of points in A .

There are many elementary consequences of the axioms underlying this definition, and we describe some of these. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

Propriedade. If $A, B \in \mathcal{F}$, then⁶ $A \setminus B \in \mathcal{F}$.

Prova. The complement of $A \setminus B$ equals $(\Omega \setminus A) \cup B$, which is the union of events and is therefore an event. Hence $A \setminus B$ is an event, by (1.3). ■

Propriedade. If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcap_{i=1}^{\infty} A_i \in \mathcal{F}$.

Prova. The complement of $\bigcap_{i=1}^{\infty} A_i$ equals $\bigcup_{i=1}^{\infty} (\Omega \setminus A_i)$, which is the union of the complements of events and is therefore an event. Hence the intersection of the A_i is an event also, as before. ■

Propriedade. If $A \in \mathcal{F}$ then $\mathbb{P}(A) + \mathbb{P}(\Omega \setminus A) = 1$.

Prova. The events A and $\Omega \setminus A$ are disjoint with union Ω , and so

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(\Omega \setminus A).$$

Propriedade. If $A, B \in \mathcal{F}$ then $\mathbb{P}(A \cup B) + \mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B)$.

Prova. The set A is the union of the disjoint sets $A \setminus B$ and $A \cap B$, and hence

$$\mathbb{P}(A) = \mathbb{P}(A \setminus B) + \mathbb{P}(A \cap B) \quad \text{by (1.14).}$$

A similar remark holds for the set B , giving that

$$\begin{aligned} \mathbb{P}(A) + \mathbb{P}(B) &= \mathbb{P}(A \setminus B) + 2\mathbb{P}(A \cap B) + \mathbb{P}(B \setminus A) \\ &= \mathbb{P}\left((A \setminus B) \cup (A \cap B) \cup (B \setminus A)\right) + \mathbb{P}(A \cap B) \quad \text{by (1.14)} \\ &= \mathbb{P}(A \cup B) + \mathbb{P}(A \cap B). \end{aligned}$$

Propriedade. If $A, B \in \mathcal{F}$ and $A \subseteq B$ then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Prova. We have that $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$. ■

It is often useful to draw a Venn diagram when working with probabilities. For example, to illustrate the formula $\mathbb{P}(A \cup B) + \mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B)$, we might draw the diagram in Figure 1.1, and note that the probability of $A \cup B$ is the sum of $\mathbb{P}(A)$ and $\mathbb{P}(B)$ minus $\mathbb{P}(A \cap B)$, since the last probability is counted twice in the simple sum $\mathbb{P}(A) + \mathbb{P}(B)$.

In the following exercises, $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space.

Exercício 7 (1.19). If $A, B \in \mathcal{F}$, show that $\mathbb{P}(A \setminus B) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$.

Exercício 8 (1.20). If $A, B, C \in \mathcal{F}$, show that

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(B \cap C) - \mathbb{P}(A \cap C) + \mathbb{P}(A \cap B \cap C).$$

⁶ $A \setminus B = A \cap (\Omega \setminus B)$ is the set of points in A which are not in B .

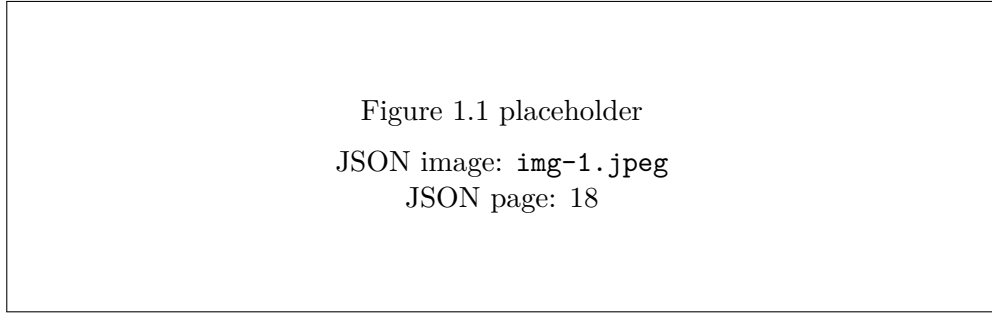


Fig. 1.1 A Venn diagram illustrating the fact that $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Exercício 9 (1.21). Let A, B, C be three events such that

$$\begin{aligned} \mathbb{P}(A) &= \frac{5}{10}, & \mathbb{P}(B) &= \frac{7}{10}, & \mathbb{P}(C) &= \frac{6}{10}, \\ \mathbb{P}(A \cap B) &= \frac{3}{10}, & \mathbb{P}(B \cap C) &= \frac{4}{10}, & \mathbb{P}(A \cap C) &= \frac{2}{10}, \\ & & \mathbb{P}(A \cap B \cap C) &= \frac{1}{10}. \end{aligned}$$

By drawing a Venn diagram or otherwise, find the probability that exactly two of the events A, B, C occur.

Exercício 10 (1.22). A fair coin is tossed 10 times (so that heads appears with probability $\frac{1}{2}$ at each toss). Describe the appropriate probability space in detail for the two cases when

- (a) the outcome of every toss is of interest,
- (b) only the total number of tails is of interest.

In the first case your event space should have $2^{2^{10}}$ events, but in the second case it need have only 2^{11} events.

5. Discrete sample spaces

Let $\mathcal{E}$ be an experiment with probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The structure of this space depends greatly on whether Ω is a countable set (that is, a finite or countably infinite set) or an uncountable set. If Ω is a countable set, we normally take $\mathcal{F}$ to be the set of all subsets of Ω , for the following reason. Suppose that $\Omega = \{\omega_1, \omega_2, \dots\}$ and, for each $\omega \in \Omega$, we are interested in whether or not this given ω is the actual outcome of $\mathcal{E}$; then we require that each singleton set $\{\omega\}$ belongs to $\mathcal{F}$. Let $A \subseteq \Omega$. Then A is countable (since Ω is countable), and so A may be expressed as the union of the countably many ω_i which belong to A , giving that $A = \bigcup_{\omega \in A} \{\omega\} \in \mathcal{F}$ by (1.4). The probability $\mathbb{P}(A)$ of the event A is determined by the collection of probabilities $\mathbb{P}(\{\omega\})$ as ω ranges over Ω , since, by (1.14),

$$\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\}).$$

We usually write $\mathbb{P}(\omega)$ for the probability $\mathbb{P}(\{\omega\})$ of an event containing only one point in Ω .

Exemplo 9 (1.23 (Equiprobable outcomes)). If $\Omega = \{\omega_1, \omega_2, \dots, \omega_N\}$ and $\mathbb{P}(\omega_i) = \mathbb{P}(\omega_j)$ for all i and j , then $\mathbb{P}(\omega) = 1/N$ for $\omega \in \Omega$, and $\mathbb{P}(A) = |A|/N$ for $A \subseteq \Omega$. $\triangle$

Exemplo 10 (1.24 (Random integers)). There are “intuitively clear” statements which are without meaning in probability theory, and here is an example: if we pick a positive integer at random, then it is an even integer with probability $\frac{1}{2}$. Interpreting “at random” to mean that each positive integer is equally likely to be picked, then this experiment would have probability space $(\Omega, \mathcal{F}, \mathbb{P})$, where

- (a) $\Omega = \{1, 2, \dots\}$,
- (b) $\mathcal{F}$ is the set of all subsets of Ω ,
- (c) if $A \subseteq \Omega$, then $\mathbb{P}(A) = \sum_{i \in A} \mathbb{P}(i) = \pi|A|$, where π is the probability that any given integer, i say, is picked.

However,

$$\begin{aligned} \text{if } \pi = 0 \text{ then } \mathbb{P}(\Omega) &= \sum_{i=1}^{\infty} 0 = 0, \\ \text{if } \pi > 0 \text{ then } \mathbb{P}(\Omega) &= \sum_{i=1}^{\infty} \pi = \infty, \end{aligned}$$

neither of which is in agreement with the rule that $\mathbb{P}(\Omega) = 1$. One possible way of interpreting the italicized statement above is as follows. Let N be a large positive integer, and let $\mathcal{E}_N$ be the experiment of picking an integer from the finite set $\Omega_N = \{1, 2, \dots, N\}$ at random. The probability that the outcome of $\mathcal{E}_N$ is even is

$$\frac{1}{2} \text{ if } N \text{ is even, and } \frac{1}{2} \left(1 - \frac{1}{N}\right) \text{ if } N \text{ is odd,}$$

so that, as $N \rightarrow \infty$, the required probability tends to $\frac{1}{2}$. Despite this sensible interpretation of the italicized statement, we emphasize that this statement is without meaning in its present form and should be shunned by serious probabilists. $\triangle$

The most elementary problems in probability theory are those which involve experiments such as the shuffling of cards or the throwing of dice, and these usually give rise to situations in which every possible outcome is equally likely to occur. This is the case of Example 1.23 above. Such problems usually reduce to the problem of counting the number of ways in which some event may occur, and the following exercises are of this type.

Exercício 11 (1.25). Show that if a coin is tossed n times, then there are exactly

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

sequences of possible outcomes in which exactly k heads are obtained. If the coin is fair (so heads and tails are equally likely on each toss), show that the probability of getting at least k heads is

$$\frac{1}{2^n} \sum_{r=k}^n \binom{n}{r}.$$

Exercício 12 (1.26). We distribute r distinguishable balls into n cells at random, multiple occupancy being permitted. Show that

- (a) there are n^r possible arrangements,

- (b) there are $\binom{r}{k}(n-1)^{r-k}$ arrangements in which the first cell contains exactly k balls,
- (c) the probability that the first cell contains exactly k balls is

$$\binom{r}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{r-k}.$$

Exercício 13 (1.27). In a game of bridge, the 52 cards of a conventional pack are distributed at random between the four players in such a way that each player receives 13 cards. Show that the probability that each player receives one ace is

$$\frac{24 \cdot 48! \cdot 13^4}{52!} = 0.105 \dots$$

Exercício 14 (1.28). Show that the probability that two given hands in bridge contain k aces between them is

$$\binom{4}{k} \binom{48}{26-k} / \binom{52}{26}.$$

Exercício 15 (1.29). Show that the probability that a hand in bridge contains 6 spades, 3 hearts, 2 diamonds and 2 clubs is

$$\binom{13}{6} \binom{13}{3} \binom{13}{2}^2 / \binom{52}{13}.$$

Exercício 16 (1.30). Which of the following is more probable:

- (a) getting at least one six with 4 throws of a die,
- (b) getting at least one double six with 24 throws of two dice?

This is sometimes called “de Méré’s paradox”, after the professional gambler Chevalier de Méré, who believed these two events to have equal probability.

6. Probability measures are continuous

There is a certain property of probability measures which will be very useful later, and we describe this next. Too great an emphasis should not be placed on the property at this stage, and we recommend to the reader that he or she omit this section at the first reading.

A sequence $A_1, A_2, \dots$ of events in a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called *increasing* if

$$A_n \subseteq A_{n+1} \quad \text{for } n = 1, 2, \dots$$

The union

$$A = \bigcup_{i=1}^{\infty} A_i$$

of such a sequence is called the *limit* of the sequence, and it is an elementary consequence of the axioms for an event space that A is an event. It is perhaps not surprising that the probability $\mathbb{P}(A)$ of A may be expressed as the limit $\lim_{n \rightarrow \infty} \mathbb{P}(A_n)$ of the probabilities of the A_n .

Teorema 11 (1.54 (Continuity of probability measures)). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. If $A_1, A_2, \dots$ is an increasing sequence of events in $\mathcal{F}$ with limit A , then

$$\mathbb{P}(A) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n).$$

We precede the proof of the theorem with an application.

Exemplo 12 (1.55). It is intuitively clear that the chance of obtaining no heads in an infinite set of tosses of a fair coin is 0. A rigorous proof goes as follows. Let A_n be the event that the first n tosses of the coin yield at least one head. Then

$$A_n \subseteq A_{n+1} \quad \text{for } n = 1, 2, \dots,$$

so that the A_n form an increasing sequence. The limit set A is the event that heads occurs sooner or later. By the continuity of $\mathbb{P}$, Theorem 1.54,

$$\mathbb{P}(A) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n).$$

However, $\mathbb{P}(A_n) = 1 - \left(\frac{1}{2}\right)^n$, and so $\mathbb{P}(A_n) \rightarrow 1$ as $n \rightarrow \infty$. Thus $\mathbb{P}(A) = 1$, giving that the probability $\mathbb{P}(\Omega \setminus A)$, that no head ever appears, equals 0. $\triangle$

Proof of Theorem 1.54. Let $B_i = A_i \setminus A_{i-1}$. Then

$$A = A_1 \cup B_2 \cup B_3 \cup \dots$$

is the union of disjoint events in $\mathcal{F}$ (draw a Venn diagram to make this clear). By (1.14),

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}(A_1) + \mathbb{P}(B_2) + \mathbb{P}(B_3) + \dots \\ &= \mathbb{P}(A_1) + \lim_{n \rightarrow \infty} \sum_{k=2}^n \mathbb{P}(B_k). \end{aligned}$$

■

However,

$$\mathbb{P}(B_i) = \mathbb{P}(A_i) - \mathbb{P}(A_{i-1}) \quad \text{for } i = 2, 3, \dots,$$

and so

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}(A_1) + \lim_{n \rightarrow \infty} \sum_{k=2}^n [\mathbb{P}(A_k) - \mathbb{P}(A_{k-1})] \\ &= \lim_{n \rightarrow \infty} \mathbb{P}(A_n) \end{aligned}$$

as required, since the last sum collapses.

The conclusion of Theorem 1.54 is expressed in terms of an *increasing* sequence of events, but the corresponding statement for a *decreasing* sequence is valid too: if $B_1, B_2, \dots$ is a sequence of events in $\mathcal{F}$ such that $B_i \supseteq B_{i+1}$ for $i = 1, 2, \dots$, then $\mathbb{P}(B_n) \rightarrow \mathbb{P}(B)$ as $n \rightarrow \infty$, where $B = \bigcap_{i=1}^{\infty} B_i$ is the limit of the B_i as $i \rightarrow \infty$. The shortest way to show this is to set $A_i = \Omega \setminus B_i$ in the theorem.

7. Problems

1. A fair die is thrown n times. Show that the probability that there are an even number of sixes is $\frac{1}{2}[1 + (\frac{2}{3})^n]$. For the purpose of this question, 0 is an even number.
2. Does there exist an event space containing just six events?
3. Prove Boole's inequality:

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \mathbb{P}(A_i).$$

4. Prove that

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) \geq 1 - n + \sum_{i=1}^n \mathbb{P}(A_i).$$

This is sometimes called *Bonferroni's inequality*, but the term is not recommended since it has multiple uses.

5. Two fair dice are thrown. Let A be the event that the first shows an odd number, B be the event that the second shows an even number, and C be the event that either both are odd or both are even. Show that A, B, C are pairwise independent but not independent.
6. Urn I contains 4 white and 3 black balls, and Urn II contains 3 white and 7 black balls. An urn is selected at random, and a ball is picked from it. What is the probability that this ball is black? If this ball is white, what is the probability that Urn I was selected?
7. A single card is removed at random from a deck of 52 cards. From the remainder we draw two cards at random and find that they are both spades. What is the probability that the first card removed was also a spade?
8. A fair coin is tossed $3n$ times. Find the probability that the number of heads is twice the number of tails. Expand your answer using Stirling's formula.
9. Two people toss a fair coin n times each. Show that the probability they throw equal numbers of heads is

$$\binom{2n}{n} \left(\frac{1}{2}\right)^{2n}.$$
10. In the circuits in Figure 1.2, each switch is closed with probability p , independently of all other switches. For each circuit, find the probability that a flow of current is possible between A and B .
11. Show that if u_n is the probability that n tosses of a fair coin contain no run of 4 heads, then for $n \geq 4$

$$u_n = \frac{1}{2}u_{n-1} + \frac{1}{4}u_{n-2} + \frac{1}{8}u_{n-3} + \frac{1}{16}u_{n-4}.$$

Use this difference equation to show that $u_8 = \frac{208}{256}$.

- * 12. Any number $\omega \in [0, 1]$ has a decimal expansion

$$\omega = 0.x_1x_2\ldots,$$

and we write $f_k(\omega, n)$ for the proportion of times that the integer k appears in the first n digits in this expansion. We call ω a *normal number* if

$$f_k(\omega, n) \rightarrow \frac{1}{10} \quad \text{as } n \rightarrow \infty$$

for $k = 0, 1, 2, \dots, 9$. On intuitive grounds we may expect that most numbers $\omega \in [0, 1]$ are normal numbers, and Borel proved that this is indeed true. It is quite another matter to exhibit specific normal numbers. Prove the number $0.1234567891011121314\dots$ is normal. It is an unsolved problem of mathematics to show that $e - 2$ and $\pi - 3$ are normal numbers also.

13. A square board is divided into 16 equal squares by lines drawn parallel to its sides. A counter is placed at random on one of these squares and is then moved n times. At each of these moves, it can be transferred to any neighbouring square, horizontally, vertically, or diagonally, all such moves being equally likely.

Let c_n be the probability that a particular corner site is occupied after n such independent moves, and let the corresponding probabilities for an intermediate site at the side of the board and for a site in the middle of the board be s_n and m_n , respectively. Show that

$$4c_n + 8s_n + 4m_n = 1, \quad n = 0, 1, 2, \dots,$$

and that

$$c_n = \frac{2}{5}s_{n-1} + \frac{1}{5}m_{n-1}, \quad n = 1, 2, \dots$$

Find two other relations for s_n and m_n in terms of c_{n-1} , s_{n-1} , and m_{n-1} , and hence find c_n , s_n , and m_n . (Oxford 1974M)

14. (a) Let $\mathbb{P}(A)$ denote the probability of the occurrence of an event A . Prove carefully, for events $A_1, A_2, \dots, A_n$, that

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) &= \mathbb{P}(A_1) - \sum_{i < j} \mathbb{P}(A_i \cap A_j) + \\ &\quad \sum_i \sum_{i < j < k} \mathbb{P}(A_i \cap A_j \cap A_k) - \dots + (-1)^{n+1} \mathbb{P}\left(\bigcap_i A_i\right). \end{aligned}$$

- (b) One evening, a bemused lodge-porter tried to hang n keys on their n hooks, but only managed to hang them independently and at random. There was no limit to the number of keys which could be hung on any hook. Alternatively, or by using (a), find an expression for the probability that at least one key was hung on its own hook.

The following morning, the porter was rebuked by the Bursar, so that in the evening she was careful to hang only one key on each hook. But she still only managed to hang them independently and at random. Find an expression for the probability that no key was then hung on its own hook.

Find the limits of both expressions as n tends to infinity.

You may assume that, for real x ,

$$e^x = \sum_{r=0}^{\infty} \frac{x^r}{r!} = \lim_{N \rightarrow \infty} \left(1 + \frac{x}{N}\right)^N.$$

(Oxford 1978M)

15. Two identical decks of cards, each containing N cards, are shuffled randomly. We say that a k -matching occurs if the two decks agree in exactly k places. Show that the probability that there is a k -matching is

$$\pi_k = \frac{1}{k!} \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^{N-k}}{(N-k)!}\right)$$

for $k = 0, 1, 2, \dots, N$. We note that $\pi_k \simeq 1/(k!e)$ for large N and fixed k . Such matching probabilities are used in testing departures from randomness in circumstances such as psychological tests and wine-tasting competitions. (The convention is that $0! = 1$.)

16. The buses which stop at the end of my road do not keep to the timetable. They should run every quarter hour, at 08.30, 08.45, 09.00, $\dots$, but in fact each bus is either five minutes early or five minutes late, the two possibilities being equally probable and different buses being independent. Other people arrive at the stop in such a way that, t minutes after the departure of one bus, the probability that no one is waiting for the next one is $e^{-t/5}$. What is the probability that no one is waiting at 09.00? One day, I come to the stop at 09.00 and find no one there; show that the chances are more than four to one that I have missed the nine o'clock bus.

You may use an approximation $e^3 \approx 20$. (Oxford 1977M)

17. A coin is tossed repeatedly; on each toss a head is shown with probability p , or a tail with probability $1 - p$. The outcomes of the tosses are independent. Let E denote the event that the first run of r successive heads occurs earlier than the first run of s successive tails. Let A denote the outcome of the first toss. Show that

$$\mathbb{P}(E \mid A = \text{head}) = p^{r-1} + (1 - p^{r-1})\mathbb{P}(E \mid A = \text{tail}).$$

Find a similar expression for $\mathbb{P}(E \mid A = \text{tail})$, and hence find $\mathbb{P}(E)$. (Oxford 1981M)

- *18. Show that the axiom that $\mathbb{P}$ is countably additive is equivalent to the axiom that $\mathbb{P}$ is finitely additive and continuous. That is to say, let Ω be a set and $\mathcal{F}$ an event space of subsets of Ω . If $\mathbb{P}$ is a mapping from $\mathcal{F}$ into $[0, 1]$ satisfying

- (i) $\mathbb{P}(\Omega) = 1, \mathbb{P}(\emptyset) = 0$,
- (ii) if $A, B \in \mathcal{F}$ and $A \cap B = \emptyset$ then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$,
- (iii) if $A_1, A_2, \dots \in \mathcal{F}$ and $A_i \subseteq A_{i+1}$ for $i = 1, 2, \dots$, then $\mathbb{P}(A) = \lim_{i \rightarrow \infty} \mathbb{P}(A_i)$,

where $A = \bigcup_{i=1}^{\infty} A_i$, then $\mathbb{P}$ satisfies

$$\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i)$$

for all sequences $A_1, A_2, \dots$ of disjoint events.

19. There are n socks in a drawer, three of which are red and the rest black. John chooses his socks by selecting two at random from the drawer, and puts them on. He is three times more likely to wear socks of different colours than to wear matching red socks. Find n .

For this value of n , what is the probability that John wears matching black socks?
(Cambridge 2008)

Referências